# Autonomous Driving Capstone Project Report

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## Executive Summary

This capstone project presents a two-part solution addressing perception and safety challenges in autonomous driving systems. Part 1 develops a vehicle classification model using transfer learning with MobileNetV2, achieving 70.92% validation accuracy across four vehicle categories. Part 2 conducts an exploratory safety analysis of 304 Tesla crash incidents to identify fatality patterns, autopilot involvement, and geographic trends. The combined approach demonstrates the integration of deep learning and data analytics for real-world autonomous driving applications.

## Problem Statement

Autonomous vehicles require accurate real-time vehicle detection for tracking, counting, and incident response systems. At the same time, increasing deployment of Tesla's Autopilot and Full Self-Driving features makes safety analysis critical. This project aims to build a robust vehicle classifier and perform a structured safety analysis of Tesla crash data.

## Part 1: Vehicle Classification Model

The dataset consisted of 4,914 labeled vehicle images. The original eight classes were consolidated into four categories (car, truck, bus, two_wheeler) to reduce class imbalance and improve model generalization. Images were resized to 224x224 pixels and split into 80% training and 20% testing sets.

Transfer learning was implemented using MobileNetV2 pretrained on ImageNet. The base model was frozen and a custom classification head was added consisting of Global Average Pooling, Dropout layers, and Dense layers with softmax activation. The model was trained using the Adam optimizer with a learning rate of 1e-4.

The model achieved a validation accuracy of 70.92%, demonstrating its suitability for real-time vehicle perception tasks in autonomous systems.

## Part 2: Tesla Autopilot Safety Analysis

The Tesla-Deaths dataset included 304 crash incidents across multiple countries. Exploratory Data Analysis (EDA) was performed to evaluate death distribution, Tesla driver fatalities, model-wise incidents, and verified autopilot deaths.

Missing values in death-related count variables were interpreted as absence of recorded fatalities and replaced with zero to maintain analytical consistency. Outliers exceeding realistic thresholds were handled during preprocessing.

Key findings indicate that 78% of incidents involved a single fatality, 38.5% involved Tesla driver deaths, and approximately 15.8% were verified Autopilot-related fatal events. The majority of incidents were recorded in the United States.

## Methodology Overview

Part 1 utilized transfer learning to reduce computational cost while leveraging pretrained feature extraction capabilities. Data preprocessing and class consolidation were applied to enhance generalization performance. Part 2 followed a structured EDA workflow including data cleaning, statistical summaries, and visualization-based trend analysis across temporal and geographic dimensions.

## Conclusions

This project successfully integrates deep learning and statistical data analysis to address both perception and safety aspects of autonomous driving systems. The vehicle classification model demonstrates practical deployment capability, while the Tesla safety analysis provides actionable insights into fatality trends and autopilot involvement. The combined findings emphasize the importance of AI-driven perception systems alongside continuous safety monitoring for responsible autonomous vehicle deployment.